

Patent Disclosure and Reliance on External Financing

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LEHIGH
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Motivation

- Innovation subsidies are massive
 - ▶ U.S. R&D tax credit
 - ▶ Patent system grants 20-year monopolies worth trillions in market value
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- A crack in that bargain: Clear disclosure has costs
 - ▶ Narrows the scope of the patent allowing greater infringement
 - ▶ Accelerates imitation
 - ▶ Lowers litigation risk for challengers
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- $\rightarrow$ Firms hide knowledge by reducing disclosure ("obfuscation")
- We ask: Can we measure obfuscation? What tradeoffs discipline firms' incentives to obfuscate? What is the role of finance in these innovation dynamics?

The paper in a slide

- **Key innovation:** We use an LLM to decompose patent complexity into a fundamental and strategic component.
 - ▶ Present stylized facts about obfuscation
 - ▶ Applicable to many settings
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Identification: Moody's 2006 adjustment to pension liabilities
 - ▶ A new mechanism for finance to shape the direction of innovation

An obfuscated patent abstract

A method, a system and a computer program product of transmitting availability information in an electronic mail system are disclosed. The system includes a first system including a sender, the first system is adapted to activate an availability indicator having a start date, wherein the activation is performed by the sender and extract availability indicator metadata from the availability indicator. The availability indicator metadata includes the start date, an end date, and at least one availability indicator message. If an e-mail has not been sent to a receiver since activation and if a current date is prior to the start date, the first system is further adapted to attach the availability indicator metadata to the e-mail, and send the e-mail to the receiver. The electronic mail system further includes a scheduling system functionally coupled to the first system, and is adapted to define the availability indicator.

Why obfuscate? Why not?

Benefit: Deter others from working in related areas

- Ambiguous language can blur the scope of a patent, acting like a “thicket”
- This has been the focus of the literature, and we will show it too

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Cost: Lower financing capacity

- Investors may struggle to understand the technology
- Deterrence also implies fewer potential buyers
- We are the first to document this effect of patent obfuscation
- Disclosure in our setting happens at the patent level, and its benefit is through the asset's financing capacity

Detecting obfuscation

- Some research has focused on the readability of the patent
- But, some technologies are fundamentally harder to describe
- How to disentangle fundamental and strategic complexity?
- Ideally, we would have a counterfactual, straightforward summary of the text to compare with
 - ▶ For example, Bushee, Gow, and Taylor (JAR 2018) compare management discussion with analyst summaries
- We use LLMs to provide this counterfactual:
Give me a simpler and clearer version of this patent abstract. Reply with JUST your version, not any label, title, header, or comments.

(prompt provided to Claude Sonnet v3.5)

The simplified patent abstract

This invention describes a way to share availability information through email. It involves a system where a sender can set up an availability indicator with start and end dates, along with a message. When the sender emails someone for the first time after setting this up, and before the start date, the system automatically adds the availability information to the email. The system also includes a scheduling feature that helps create these availability indicators.

The simplified patent abstract

This invention describes a way to share availability information through email. It involves a system where a sender can set up an availability indicator with start and end dates, along with a message. When the sender emails someone for the first time after setting this up, and before the start date, the system automatically adds the availability information to the email. The system also includes a scheduling feature that helps create these availability indicators.

- Patent number 9547842, granted to IBM.
- Often described as a general patent on auto-reply features.
The simplified language shows it is actually narrower than that.
- Gunning-Fog complexity improved by 7 grade levels from 19.71 (post-graduate) to 12.81 (college freshman)

Quantifying patent abstract complexity

- Scale this to all 2.6m patents matched to public firms, 1976-2023
- For each patent, we prompt the LLM to rewrite the abstract
 - ▶ Focus on abstracts to keep costs manageable:
If a filer obfuscates anywhere, they should obfuscate everywhere.
 - ▶ High cosine similarity of embeddings between original and rewrite.
- Quantify the complexity of both the original abstract, and the LLM rewrite
 - ▶ Presenting Gunning-Fog results today
- For any linguistic complexity measure $f(\cdot)$,

$$\text{Obfuscation} = f(\text{Original abstract}) - f(\text{LLM-simplified abstract}) \quad (1)$$

$$= \text{Total Complexity} - \text{Fundamental Complexity} \quad (2)$$

Example: A research abstract

Before: We decompose patent complexity into fundamental and strategic components using LLMs, interpreting the latter as obfuscation. Patent obfuscation has risen sharply across all major technology areas over the last 15 years. While obfuscation deters competitor innovation and reduces infringement litigation, it also decreases patents' financing capacity. Using IV tests, we show obfuscated patents are pledged as collateral and sold less frequently. In difference-in-differences tests, firms that recently pledged patents or experienced negative credit-rating shocks reduce obfuscation and policy shocks affecting obfuscation have weaker effects for financially constrained firms. Our findings highlight new connections between financing and the path of innovation.

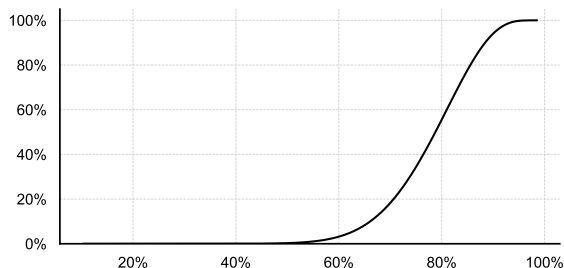
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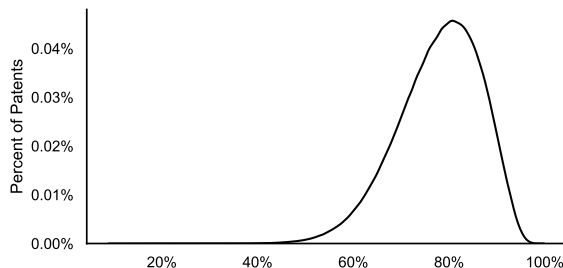
After: Patents have become harder to understand over the past 15 years, but not just because the technology is more complex. Companies are deliberately making patents more difficult to read—a practice we call obfuscation. This strategy helps deter competitors and reduces patent disputes, but it comes with a cost: banks and investors are less willing to accept obfuscated patents as collateral for loans or to purchase them. We confirm this tradeoff by showing that companies needing financing write clearer patents, and that financially healthy firms are more likely to use obfuscation. These findings reveal that a company's financial situation shapes how it writes patents and influences the direction of technological innovation.

Gunning-Fog improved by 2 grade levels, from 20 (A.B.D.) to 18 (M.S.).

Rewritten abstracts maintain meaning of original abstracts



Empirical CDF of Cosine Similarity



KDE of Cosine Similarity

Embeddings from *PatentSBERTa* (Bekamiri et al., 2024).

Different LLMs and repeated runs yield similar scores

Random sample of 1,000 patents:

Model	Legend	(1)	(2)	(3)	(4)	(5)	(6)	(7)
anthropic/claude-3-5	(1)							
anthropic/claude-3-7-sonnet-latest	(2)	0.97						
anthropic/claude-4-opus	(3)	0.96	0.94					
google/gemini-2.5-flash-lite	(4)	0.94	0.95	0.91				
meta/Llama-3.3-70B-Instruct-Turbo	(5)	0.92	0.93	0.90	0.91			
openai/gpt-4o	(6)	0.98	0.98	0.94	0.95	0.93		
xai/grok-4-1-fast-non-reasoning	(7)	0.96	0.97	0.93	0.94	0.92	0.97	

We also rerun the LLM simplification 5 times and find 99%+ correlations.

Fact 1: + with info. asymmetry and - in trade secrets

	Info Asymmetry Percentile			Trade Secret (Boolx100)		
	(1)	(2)	(3)	(4)	(5)	(6)
Obfuscation (GF) (z)	0.423*** (0.09)	0.658*** (0.17)	0.410*** (0.09)	-0.338** (0.15)	-0.671** (0.33)	-0.320** (0.16)
Total Complexity (GF) (z)		-0.245 (0.18)			0.348 (0.36)	
Fundamental Complexity (GF) (z)			-0.060 (0.04)			0.085 (0.09)
Fixed effects:						
CPC-year	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	1,802,286	1,802,286	1,802,286	1,455,104	1,455,104	1,455,104
R ²	0.49	0.49	0.49	0.27	0.27	0.27

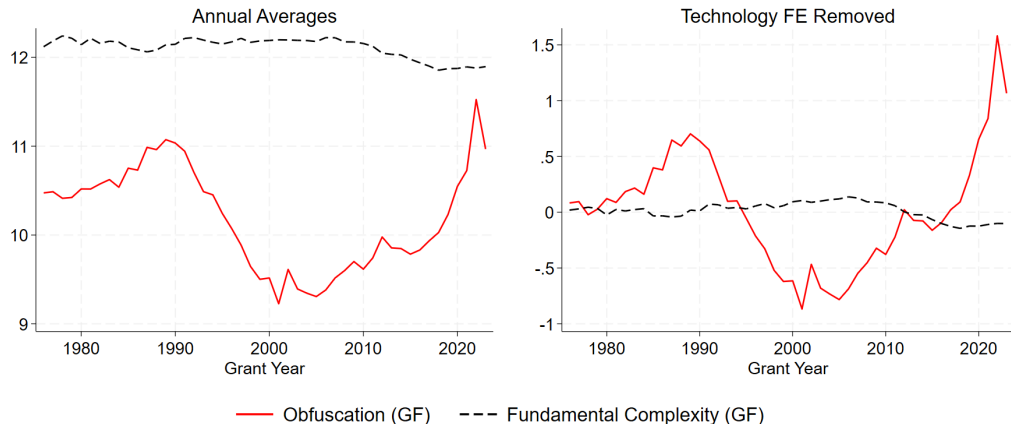
- Information asymmetry: Validation of weaker disclosure
- Information asymmetry: Horse race w.r.t. total complexity

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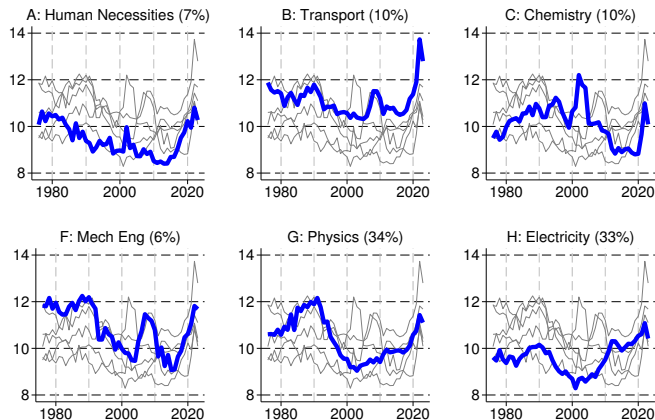
- Trade secrets: A substitute for alternative IP protection

Fact 2: Obfuscation fell during the 1990s but has steadily risen since



- Obfuscation drives variation in Total complexity/Patent complexity

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- By technology area: Physics and electricity drive aggregate pattern
- Substantial variation; Rank order changes over time

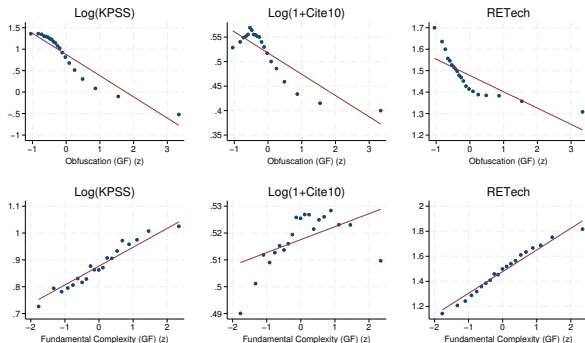
Fact 3: Correlations with patent impact and disruption/technology waves

Obfuscation (top row):

- (-) Follow-on innovation
- (-) Patent value (Kogan et al. 2017)
- (-) Rapidly evolving tech areas

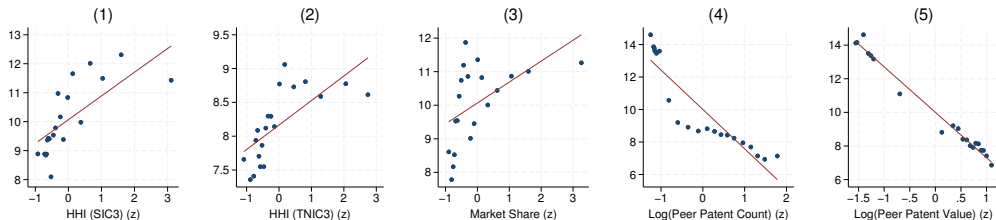
Fundamental Comp. (bottom row):

- (+) Follow-on innovation
- (U) Patent value (Kogan et al. 2017)
- (+) Rapidly evolving tech areas



→ Illustrates the value of the decomposition we propose

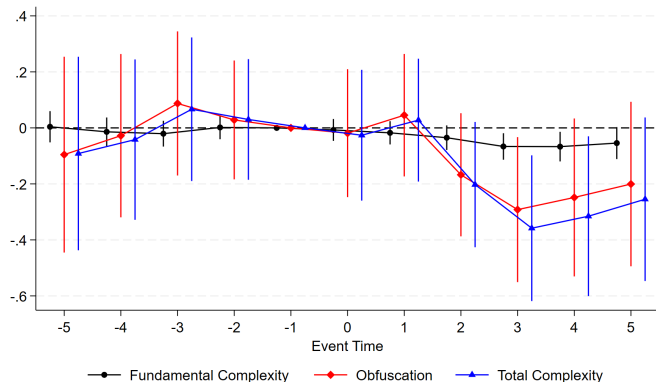
Fact 4: Obfuscation is highly related to strategic factors



→ **Obfuscation increases as competitive threats rise (1-3)**

→ **Obfuscation falls as the value of deterrence drops (4-5)**

Fact 5: Obfuscation declines after firms start pledging patents



- Event study around the first pledge made by a firm
- Which increases the financial salience of (new) patents
- Hypothesis: Firms respond by writing clearer patents
- Result: Drop in complexity driven by obfuscation decline

→ **Suggests a new financial factor disciplining the strategic tradeoff of obfuscation.**

Summary of New Facts about Patent Obfuscation

- ① Steadily rising since 2000, with possible consequences for innovation and growth
- ② Not driven by fundamental technology shocks
- ③ Large variation across technologies and time

Summary of New Facts about Patent Obfuscation

- ➊ Steadily rising since 2000, with possible consequences for innovation and growth
- ➋ Not driven by fundamental technology shocks
- ➌ Large variation across technologies and time
- ➍ + related to information asymmetry, - to trade secrets
- ➎ Negatively related to patent impact (deterrence)
- ➏ Rises with competitive threats, falls as deterrence value drops
- ➐ Falls when salience of financial capacity of patents rises

Rest of paper: Evidence on financial/strategic tradeoff

- ➊ Above suggests $\Rightarrow$ Obfuscation carries a novel cost: weaker access to external financing
- ➋ Now we look for clean evidence on this tradeoff
- ➌ Four shocks to ability and incentives to obfuscate
- ➍ Each approach as strength and weakness
- ➎ Jointly: a compelling case that financial constraints discipline obfuscation

Description of identification strategies

- Strategy 1: IV using random assignment of patent examiners
 - ▶ Some examiners enforce greater clarity (Dyer et al., JAE 2024)
- Strategy 2: *Nautilus v. Biosig* Supreme Court decision
 - ▶ Articulated a stronger uniform requirement for clarity
- Strategy 3: American Inventors Protection Act (AIPA)
 - ▶ Patents filed after 2000 had to be published 18 months after filing *unless US-only*
- Strategy 4: Moody's Adjustment to Pension Calculations
 - ▶ Change in treatment of underfunded pensions reduced adjusted leverage and positively shocked ratings (Kraft 2015, Kisgen 2019)

IV Validation: Strict examiners force simpler language

Outcome: Does abstract become more readable during review (between *application* and *grant*)

Rej Rates Are:	Examiner Only		Unit-Adjusted	
	(1)	(2)	(3)	(4)
Rej Rates For:	Apps.	Grants	Apps.	Grants
Examiner 112 Strictness (z)	0.647*** (0.06)	0.639*** (0.06)	0.470*** (0.04)	0.468*** (0.04)
Examiner 101 Strictness (z)	0.764*** (0.07)	0.763*** (0.07)	0.359*** (0.03)	0.360*** (0.03)
Examiner 102 Strictness (z)	0.202*** (0.04)	0.201*** (0.04)	0.175*** (0.04)	0.175*** (0.04)
Examiner 103 Strictness (z)	0.157*** (0.04)	0.157*** (0.04)	0.152*** (0.04)	0.152*** (0.04)
Fundamental Complexity (GF) (z)	0.082** (0.04)	0.082** (0.04)	0.082** (0.04)	0.082** (0.04)
Fixed effects:				
CPC-year	Y	Y	Y	Y
Art-unit-year	Y	Y	Y	Y
Obs.	395,304	395,303	395,273	395,272
R ²	0.06	0.06	0.06	0.06

Instrumental variables results

	1st Stg	Log of 1 plus cites received from			Other outcomes		
	(1) Obf. (z)	(2) All	(3) Others	(4) Self	(5) Infringed	(6) PatSold	(7) Pledged
Ex. 112 Strictness (z)	-0.010*** (0.00)						
$\widehat{Obfuscation}(z)$		-2.162*** (0.47)	-1.273*** (0.32)	-1.515*** (0.33)	-0.046*** (0.02)	-0.381*** (0.12)	-0.202** (0.08)
Ex. Other Strictness (z)	-0.009*** (0.00)	-0.006 (0.00)	-0.005* (0.00)	-0.002 (0.00)	-0.000 (0.00)	-0.001 (0.00)	0.001 (0.00)
Fund. Complexity (z)	-0.200*** (0.00)	-0.409*** (0.09)	-0.236*** (0.06)	-0.292*** (0.07)	-0.009*** (0.00)	-0.074*** (0.02)	-0.039** (0.02)
Fixed effects:							
CPC-year	Y	Y	Y	Y	Y	Y	Y
Art-unit-year	Y	Y	Y	Y	Y	Y	Y
Obs.	733,013	733,013	733,013	733,013	733,013	733,013	733,013
Cragg-Donald F	27.71						

- Strategic deterrence: Obfuscated patents $\implies$ less follow-on innovation.
- Financial tradeoff: But also, less liquidity and collateral usage.

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Examiner IV strategy: Takeaways

- Examiners can influence the degree of patent obfuscation
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- But they also are less likely to be sold or pledged
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- Obfuscation tradeoff: strategic deterrence vs. financing capacity
 - ▶ max Obfuscation may *not* be optimal for firms with greater external financing needs
 - ▶ Supports the conclusions from pledging event-time analysis
 - ▶ We next test this prediction with exogenous shocks to obfuscation ability

Nautilus v. Biosig Shock

Z variable:	(1) None	(2) Pledged	(3) HighLev	(4) None	(5) Pledged	(6) HighLev
Exposure	1.214*** (0.03)	1.072*** (0.06)	1.242*** (0.03)			
Post	1.478*** (0.33)	3.668*** (0.72)	2.109*** (0.46)	0.420** (0.21)	1.488*** (0.45)	0.830*** (0.19)
Exposure x Post	-0.158*** (0.05)	-0.512*** (0.12)	-0.271*** (0.08)	-0.071** (0.03)	-0.254*** (0.07)	-0.138*** (0.03)
Z		-0.226 (0.17)	0.967*** (0.36)			-0.509 (0.41)
Z x Exposure		0.061** (0.03)	-0.081 (0.05)			
Z x Post		-1.016*** (0.28)	-1.767*** (0.62)		-0.455** (0.18)	-1.105*** (0.36)
Z x Exposure x Post		0.163*** (0.05)	0.292*** (0.10)		0.078*** (0.03)	0.174*** (0.06)
Fixed effects:						
Year	N	N	N	Y	Y	Y
Firm	N	N	N	Y	Y	Y
Obs.	503,093	503,093	503,086	503,093	503,093	503,086
R ²	0.12	0.12	0.12	0.14	0.14	0.14

- Increases requirement for patent clarity
 - Ashtor (2022) documents a significant increase in §112(b) rejections

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 - Ashtor (2022) documents a significant increase in §112(b) rejections
- Exposure = above median obfuscation pre-event
- Here: Obfuscation falls for exposed firms

Nautilus v. Biosig Shock

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Year	N	N	N	Y	Y	Y
Firm	N	N	N	Y	Y	Y
Obs.	503,093	503,093	503,086	503,093	503,093	503,086
R ²	0.12	0.12	0.12	0.14	0.14	0.14

- Increases requirement for patent clarity
 - Ashtor (2022) documents a significant increase in §112(b) rejections
- Exposure = above median obfuscation pre-event
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Nautilus v. Biosig Shock

	(1)	(2)	(3)	(4)	(5)	(6)
Z variable:	None	Pledged	HighLev	None	Pledged	HighLev
Exposure	1.214*** (0.03)	1.072*** (0.06)	1.242*** (0.03)			
Post	1.478*** (0.33)	3.668*** (0.72)	2.109*** (0.46)	0.420** (0.21)	1.488*** (0.45)	0.830*** (0.19)
Exposure x Post	-0.158*** (0.05)	-0.512*** (0.12)	-0.271*** (0.08)	-0.071** (0.03)	-0.254*** (0.07)	-0.138*** (0.03)
Z		-0.226 (0.17)	0.967*** (0.36)			-0.509 (0.41)
Z x Exposure		0.061** (0.03)	-0.081 (0.05)			
Z x Post		-1.016*** (0.28)	-1.767*** (0.62)		-0.455** (0.18)	-1.105*** (0.36)
Z x Exposure x Post		0.163*** (0.05)	0.292*** (0.10)		0.078*** (0.03)	0.174*** (0.06)
Fixed effects:						
Year	N	N	N	Y	Y	Y
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AIPA Shock

	(1)	(2)	(3)	(4)	(5)	(6)
Z variable:	None	Pledged	HighLev	None	Pledged	HighLev
Exposure	5.312*** (0.381)	1.220 (0.825)	4.260*** (0.483)			
Post	-0.541* (0.311)	-1.361*** (0.439)	-0.801** (0.374)			
Exposure x Post	1.266** (0.543)	4.200*** (1.168)	2.904*** (0.700)	1.192*** (0.238)	0.755 (0.579)	1.621*** (0.281)
Z		-0.856* (0.449)	0.479 (0.481)			-0.607* (0.313)
Z x Exposure		4.497*** (0.919)	1.753*** (0.648)			0.750* (0.416)
Z x Post		0.861 (0.553)	1.045* (0.597)		0.529 (0.386)	0.740** (0.325)
Z x Exposure x Post		-2.930** (1.308)	-3.842*** (0.963)		0.478 (0.629)	-1.111** (0.441)
Fixed effects:						
Year	N	N	N	Y	Y	Y
Firm	N	N	N	Y	Y	Y
Obs.	982,497	982,497	976,986	982,496	982,496	976,985
R ²	0.02	0.02	0.02	0.10	0.10	0.10

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- → Highlights the strategic vs financing tradeoff

Moody's Shock: Update to pension liability adjustment method

	(1) Obfuscation	(2) Fundamental Complexity
Underfunded $\times$ Post	0.584** (0.25)	-0.043 (0.04)
Fixed effects:		
CPC-year	Y	Y
Firm	Y	Y
Obs.	151,618	151,618
R^2	0.05	0.08

- Shock to financial constraints (instead of disclosure)
- Treatment firms get positive credit rating shocks
- *Underfunded* = large underfunded pensions
 - ▶ Inspired by Kraft (2015) and Kisgen (2019)
- Obfuscation rises for patents of treated firms

Summary of results of four identification strategies

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Overall: → Financial constraints discipline obfuscation

Conclusions and takeaways

- A novel approach to measuring obfuscation using LLMs and standard tools from linguistics.
 - ▶ Plausibly applicable in many disclosure settings
- Highlight new tradeoff: patent obfuscation can hinder the patent's financing capacity
 - ▶ In other settings, disclosure is recognized as a tradeoff at the firm level
Our setting is unique for representing disclosure about a specific asset
- Financing constraints can motivate greater patent disclosure, and increase knowledge spillovers
 - ▶ A new mechanism for finance to shape the direction of innovation

Thank you!

Questions?

Talk to and listen to my papers at <https://bowen.finance>